
MercMinds

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MercMinds	
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Abstract

The Predictive Insights Hackathon required participating teams to predict the probability that individuals in a Round 9 cohort would be employed using longitudinal survey data. MercMinds approached the problem as a binary classification task and focused on developing an ensemble-based modelling strategy capable of capturing both linear and non-linear relationships within the data. The modelling process began with a Logistic Regression baseline, achieving a Public AUC of 0.59229, and progressively introduced tree-based models, neural networks, regularised linear models, latent-factor representations and ensemble optimisation.

A 64-32 Multi-Layer Perceptron (MLP) substantially improved performance to 0.64453, while combining MLP and Logistic Regression predictions increased the Public AUC to 0.65089. Further improvements were achieved through out-of-fold blending, regularisation, multiple random seeds and latent-factor modelling. The best-performing approach was a Non-Negative Least Squares (NNLS) Super-Learner, achieving a Public AUC of 0.66054. Simplex optimisation reproduced this peak, while the Grand Master Stacker provided a near-identical alternative. In contrast, AutoGluon demonstrated weaker generalisation, with a Public AUC of 0.63622 despite strong internal cross-validation performance.

Overall, MercMinds improved the Public AUC by 0.06825 points, demonstrating that model diversity, regularisation, robust feature representations and constrained ensemble optimisation were more effective than relying on a single predictive algorithm.

1. Introduction

Youth unemployment remains a significant socioeconomic challenge in South Africa, particularly among young people who face barriers to entering and remaining in the labour market. Understanding the factors associated with employment can provide valuable insights into labour-market participation and support the development of more effective policies and interventions. However, employment outcomes are influenced by a combination of demographic, educational, geographic and socioeconomic factors, making them difficult to predict using individual characteristics alone.

Our project addresses this challenge through a machine-learning approach using longitudinal youth survey data. The dataset tracks participants across multiple survey rounds, providing historical information about their demographic characteristics, education, geographic location and labour-market experiences. This longitudinal structure allows the project to examine patterns in participants' previous experiences and use these patterns to predict their employment status in Round 9.

The primary objective of the project is to develop a predictive model that estimates the probability of each participant being employed in Round 9. This is formulated as a binary classification problem, where the outcome represents whether a participant is employed or not. The modelling process involves exploratory data analysis, data preprocessing, feature engineering and the comparison of different machine-learning algorithms to identify an effective predictive approach.

Model performance is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC), which is appropriate for the challenge because it evaluates how effectively the model ranks participants according to their likelihood of employment. Rather than focusing only on whether individual predictions are classified correctly, AUC measures the model's ability to distinguish between participants who are employed and those who are not.

Through this approach, MercMinds aims to demonstrate how machine learning and longitudinal survey data can be used to generate predictive insights into youth employment outcomes in South Africa. The project focuses not only on achieving strong predictive performance but also on developing a structured and reproducible modelling workflow that can provide meaningful insights into the factors and historical patterns associated with employment.

2. Problem Statement

The Predictive Insights Hackathon provides participants with a longitudinal dataset containing demographic, educational, geographic, socioeconomic and employment-related information collected across multiple survey rounds. The objective is to use the available historical

information to predict the probability that each participant in the Round 9 cohort will be employed.

The prediction task presents several modelling challenges. First, employment is a binary outcome influenced by multiple interacting characteristics rather than a single explanatory variable. Second, the longitudinal structure of the dataset means that participants may have different amounts of historical information available. Third, the dataset contains missing observations across a number of variables, requiring appropriate treatment during preprocessing and feature engineering. Finally, because the competition evaluates predicted probabilities using AUC, a model must rank participants effectively rather than simply maximise classification accuracy at a fixed threshold.

MercMinds therefore investigated the following question:

“How can demographic, educational, geographic, work-readiness and longitudinal employment information be transformed into predictive features and combined within a machine-learning model to maximise the AUC of Round 9 employment predictions?”

The team's approach focuses on developing a reproducible modelling pipeline, evaluating alternative algorithms and feature representations, and testing model stability using historical survey rounds before generating the final Round 9 predictions.

3. Methods

Our solution followed an iterative machine-learning methodology consisting of data exploration, preprocessing, domain-specific feature engineering, model development, cross-validation, latent feature extraction, and ensemble optimisation. A total of 100 experiments were conducted to identify an approach that could generalise effectively to the unseen Round 9 evaluation cohort.

Data Preparation and Feature Engineering

The training data contained 16,719 observations, with 16,653 valid employment-status targets. The test data contained 3,230 observations.

The data was examined for missing values, categorical variables, numerical variables and high-cardinality text fields. Suitable categorical variables were one-hot encoded, while some high-cardinality fields were removed because they caused overfitting.

Missing values in historical variables, such as previous employment and survey timing, were treated as meaningful indicators. Features such as ‘is_first_time’ were created to identify participants without previous survey records.

Additional features were engineered to represent labour-market history, academic performance, education, geography and demographic characteristics. Matric percentage bands were converted

into approximate numerical values, and composite features such as mathematics and STEM scores were created.

Models Tested

Logistic Regression was used as the baseline model. Other tested approaches included:

- Regularised linear and ElasticNet models
- Random Forest and Extra Trees
- Gradient-boosting models
- Multi-Layer Perceptrons (MLPs)
- CatBoost
- AutoGluon
- Ensemble and stacking methods

Different MLP architectures and random seeds were tested to improve model stability.

Validation and Latent-Factor Modelling

Stratified five-fold cross-validation was used to evaluate model performance and generate out-of-fold predictions. Temporal validation was also considered because the model needed to predict a future survey round.

Latent factors were created to summarise related variables into three main areas:

1. Academic Human Capital
2. Labour Market Momentum
3. Socio-Economic Readiness

Factor Analysis and Partial Least Squares were used to create these lower-dimensional representations, which were then included in the modelling pipeline.

Ensemble Modelling

The final solution combined the predictions of several strong models:

- Quantile ElasticNet
- Deep MLP
- Medium MLP
- Standard ElasticNet SAGA

This combination allowed the model to use both the stability of linear models and the ability of neural networks to capture non-linear relationships.

The ensemble weights were optimised using simplex-constrained learning, meaning that all weights had to be non-negative and sum to one. The final weights were approximately 43.8% Quantile ElasticNet, 25.5% Deep MLP, 19.8% Medium MLP and 10.9% Standard ElasticNet. This approach performed better than unconstrained stacking because it avoided negative weights and unstable predictions.

4. Results

A total of 100 experiments were conducted during the development of the MercMinds solution. The experiments progressively investigated different representations, algorithms, validation strategies and ensemble techniques. The initial Logistic Regression baseline achieved a public leaderboard AUC of 0.59229. Through iterative improvements, the final solution achieved a peak public leaderboard AUC of **0.66054**, representing an improvement of **0.06825 AUC points** over the baseline.

Performance Progression

The major stages of the modelling process are summarized below:

Experiment	Method	OOF AUC	Public AUC	Key Finding
Exp 30	Logistic Regression baseline	0.5982	0.59229	Established the baseline
Exp 31	Extra Trees	0.6215	0.61732	Tree-based approach improved over baseline but remained limited
Exp 33	MLP 64-32	0.6481	0.64453	Non-linear modelling produced a substantial improvement
Exp 35	MLP + Logistic Regression	0.6542	0.65089	Combining linear and neural predictions improved performance
Exp 37	5-Fold OOF blending	0.6616	0.65247	Cross-validation reduced prediction variance
Exp 44	Regularised linear ensemble	0.6621	0.65630	ElasticNet-based models provided strong generalisation
Exp 51	Multi-seed linear + neural ensemble	0.6634	0.65693	Multiple seeds improved ensemble stability
Exp 59	SEM latent-factor ensemble	0.6648	0.65832	Latent representations improved performance
Exp 68	NNLS Super-Learner	0.6668	0.66054	Non-negative ensemble optimisation achieved the peak

Exp 71	Grand Master Stacker	0.6669	0.66051	Provided a near-identical safety alternative
Exp 93	Simplex SLSQP optimisation	0.6669	0.66054	Reproduced the peak score
Exp 96	AutoGluon	0.6684	0.63622	High CV performance failed to transfer to Round 9
Exp 100	25-Fold Simplex ensemble	0.6672	0.66029	Confirmed the stability of the final modelling approach

The initial Logistic Regression model achieved a public AUC of **0.59229**, providing a baseline for comparison. Performance improved significantly when non-linear models were introduced, with the 64-32 MLP achieving **0.64453** and a combined MLP-Logistic Regression model reaching **0.65089**.

Further improvements came from five-fold cross-validation, out-of-fold blending, regularised linear models, and multi-seed ensembles. These methods increased the public AUC to **0.65693** by improving prediction stability and combining complementary models.

Latent-factor modelling produced an additional improvement by reducing the effect of correlated survey variables and representing concepts such as academic ability, labour-market experience and socioeconomic readiness more effectively. This approach achieved an AUC of **0.65832**.

The best result was obtained using a non-negative super-learner ensemble, which achieved a public AUC of **0.66054**. The final ensemble combined Quantile ElasticNet, two MLP models and Standard ElasticNet, showing that combining regularised linear and neural models was more effective than relying on one algorithm.

Some methods were underperformed. Tree-based models achieved lower leaderboard scores, while AutoGluon showed signs of overfitting by achieving a high internal AUC but a much lower public AUC. Unconstrained stacking and pseudo-labelling also reduced performance.

Overall, the model improved from **0.59229 to 0.66054**, an absolute increase of **0.06825 AUC points**. The main improvements came from better feature representation, regularisation, model diversity and constrained ensemble optimisation.

5. Discussion and Model Interpretation

The experiment results demonstrate that the performance of the MercMinds solution improved substantially through changes to feature representation, model selection, validation and ensemble optimisation. However, predictive performance alone does not explain which characteristics were most important in determining employment predictions. To better understand the behaviour of the final model, feature coefficients, latent-factor loadings and non-linear patterns learned by the models were examined.

Feature Importance and Variable Contributions

The strongest predictive signals came from previous labour-market participation, followed by academic performance, education, geographic location and work-readiness indicators.

Labour Market Persistence

‘employed_lag’, ‘tenure_lag’ and ‘days_since_last_obs’ were important predictors. Previous employment and longer employment tenure were positively associated with future employment predictions, while longer periods between observations were associated with lower predicted employment. These variables formed part of the Labour Market Momentum latent factor.

Matric and STEM Performance

Pure Mathematics, Physical Science and the engineered ‘stem_composite’ feature contributed positively to employment predictions. Pure Mathematics provided more predictive information than Mathematical Literacy after other educational variables were considered. This reflects an association in the dataset rather than evidence of direct employer screening.

Education, School Quintile and Work Readiness

School quintile and ‘work_readiness_score’ provided additional predictive information. Higher school quintiles and stronger work-readiness scores were generally associated with higher employment predictions. The neural networks also captured interactions between these characteristics, although these relationships should not be interpreted as causal.

Geographic Distribution

‘province’, ‘district’ and ‘municipality’ contributed to the model, with Gauteng and Western Cape generally showing stronger positive signals than Eastern Cape and Limpopo. These differences may reflect variation in access to economic centres, transport, education and local labour demand.

Underlying Socio-Economic and Labour Market Patterns

The feature importance analysis identified several patterns associated with employment predictions. These findings help explain the model but should not be interpreted as causal relationships.

Labour Market Persistence and Possible Scarring

‘employed_lag’ was the strongest labour-market predictor, showing that employment status tends to persist between survey rounds. Longer periods since the previous observation were associated with lower employment predictions, possibly reflecting reduced work experience or weaker labour-market connections.

Academic Performance

Pure Mathematics and Physical Science provided useful predictive information. Their stronger contribution may reflect links to further education, technical occupations, or fields with higher labour demand. However, the model identifies associations rather than proving that these subjects directly cause higher employment.

Geographic Differences

Gauteng and Western Cape produced stronger positive employment signals than some other provinces. This may reflect differences in economic activity, labour demand, and access to employment opportunities. As a result, participants with similar qualifications may receive different predictions based on their location.

Socioeconomic and Work-Readiness Factors

School quintile and work readiness both contributed to employment predictions. This suggests that employment outcomes are influenced by a combination of individual preparation and socioeconomic context, rather than by individual ability alone.

First-Time Participants

The ‘is_first_time’ variable identified participants without previous survey observations. These participants provided less historical information, increasing prediction uncertainty. This highlights the importance of treating missing longitudinal information carefully, as the absence of previous observations may itself contain useful information.

Policy Implications

The patterns identified by the model suggest several areas that could be considered when designing interventions aimed at improving youth employment outcomes.

Empirical Finding	Possible Interpretation	Potential Intervention
Previous employment status was the strongest labour-market predictor.	Employment outcomes show strong persistence between survey rounds.	Provide early intervention and re-employment support following job loss or entry into unemployment.
Longer periods between observations were associated with lower employment predictions.	Extended periods outside the labour market may be associated with weaker employment prospects.	Develop rapid re-engagement, job-search and placement programmes for unemployed youth.
Pure Mathematics and STEM-related performance provided strong predictive signals.	Academic pathways may be associated with different employment and further-education opportunities.	Provide bridging programmes, quantitative skills development and technical training opportunities.
Metropolitan provinces produced stronger employment signals.	Employment opportunities may be concentrated around major economic centres.	Consider transport support, decentralised employment services and remote-work opportunities for youth in underserved regions.
Work readiness was positively associated with employment predictions.	Employment preparation and behavioural readiness contain useful predictive information.	Expand career-readiness, interview preparation, workplace-skills and job-placement programmes.
First-time participants had less historical information available.	Limited employment history increases uncertainty when predicting future outcomes.	Strengthen support and screening mechanisms for young people entering the labour market for the first time.

These implications should not be interpreted as direct recommendations produced by the model. Rather, they represent potential areas for further investigation based on the relationships identified in the predictive analysis.

Overall Interpretation

Employment prediction was driven by a combination of labour-market history, academic performance, socioeconomic context and geographic factors. Previous employment was the strongest predictor, while academic, STEM-related and geographic variables provided additional information.

The MercMinds modelling strategy combined linear models, neural networks, latent factors and ensemble optimisation to capture both simple and complex relationships. The final AUC of

0.66054 therefore reflects not only improved predictive performance but also useful insight into the factors associated with youth employment outcomes.

6. Conclusion

This paper presented the modelling approach developed by MercMinds for the Predictive Insights Hackathon, where the objective was to predict the probability that participants in the Round 9 cohort would be employed. As all participating teams were given the same prediction problem and evaluated using AUC, the primary focus of the project was to identify modelling techniques that could extract useful signals from the longitudinal survey data and generalise effectively to the Round 9 evaluation set.

MercMinds followed an iterative experimental approach, beginning with a Logistic Regression baseline and progressively introducing nonlinear models, regularisation, cross-validation and ensemble techniques. The initial Logistic Regression model achieved a public AUC of 0.59229, establishing a baseline against which subsequent approaches could be evaluated. Extra Trees improved performance to 0.61732, while the introduction of a 64-32 Multi-Layer Perceptron produced a substantial improvement to 0.64453. Combining MLP and Logistic Regression predictions further increased the public AUC to 0.65089, demonstrating the benefit of combining models with different predictive characteristics.

Further experiments focused on improving prediction stability and reducing model-specific variance. Five-fold out-of-fold blending increased the public AUC to 0.65247, while regularised linear ensembles and multi-seed linear-neural ensembles increased performance further to 0.65630 and 0.65693 respectively. These results indicated that model diversity, regularisation and out-of-fold prediction were important components of the MercMinds modelling strategy.

The introduction of latent-factor modeling provided an additional improvement. The SEM latent-factor ensemble achieved a public AUC of 0.65832, suggesting that representing correlated survey variables through broader latent concepts could provide useful predictive information. This was followed by constrained ensemble optimisation, where the NNLS Super-Learner achieved the highest public AUC of 0.66054. The ensemble combined Quantile ElasticNet, Standard ElasticNet and two MLP models, demonstrating that complementary linear and nonlinear predictions could outperform the individual models from which they were constructed. Simplex SLSQP optimisation independently reproduced the same public AUC of 0.66054, providing further evidence that the peak result was not dependent on a single optimisation implementation.

Not all approaches that performed strongly during internal validation transferred successfully to the public evaluation. AutoGluon, for example, achieved the highest OOF AUC observed in the experiments at 0.6684, but its public AUC decreased substantially to 0.63622. This difference highlights the risk of relying exclusively on internal cross-validation performance when the ultimate evaluation is performed on a separate Round 9 cohort. Similarly, experiments involving unconstrained stacking and pseudo-labelling did not provide improvements over the constrained ensemble approach.

The findings demonstrate the value of systematic experimentation in a competitive machine-learning setting. In particular, the difference between internal OOF performance and public leaderboard performance reinforces the importance of evaluating model stability and generalisation rather than simply selecting the model with the highest cross-validation score. Future work could investigate improved longitudinal feature representations, alternative latent-factor structures, more robust ensemble validation strategies and additional methods for reducing the gap between internal validation performance and Round 9 generalisation.

Code

- *Code repository:* https://github.com/entisaar-elfadl/PI_Hackathon_MercMinds

References

Predictive Insights Hackathon: Youth Employment Predication Dataset(Competition Data)